JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Basic Algebra (2005–2026 Archive)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
3	6 M	15 Min	MCQ • MSQ • NAT (2005–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Select (MSQ)	1	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	2	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q60 • ID: JAM_2026_Q60) NAT +2 M Neg: Nil (0)

Q.60	Let $\binom{n}{r}$ denote the number of ways of choosing r distinct objects out of n distinct objects. Then, $$\binom{5}{0}+\binom{6}{1}+\binom{7}{2}+\binom{8}{3}+\binom{9}{4}+\binom{10}{5}+\binom{11}{6} = \_\_\_\_\_.$$

Question 2 (JAM 2026 • Q58 • ID: JAM_2026_Q58) NAT +2 M Neg: Nil (0)

Q.58	A fruit shop has 4 different types of bananas. The number of ways in which 12 bananas can be bought with at least one banana from each type, is ______.

Question 3 (JAM 2026 • Q38 • ID: JAM_2026_Q38) MSQ +2 M Neg: Nil (0)

Q.38	Let $\binom{n}{r}$ denote the number of ways of choosing r distinct objects out of n distinct objects. Which of the following statements is/are TRUE?
(A)	$\sum_{k=0}^{6}(-1)^k\binom{13}{2k} = -64.$
(B)	$\sum_{k=0}^{6}(-1)^k\binom{13}{2k+1} = 128.$
(C)	$\sum_{k=0}^{6}(-1)^k\binom{13}{2k} = 64.$
(D)	$\sum_{k=0}^{6}(-1)^k\binom{13}{2k+1} = -128.$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Basic Algebra (2005–2026 Archive) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	**924 to 924**
2	NAT	+2	**165 to 165**
3	MSQ	+2	**A**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.